

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	C Prathibha	Reg. No.:	192424293
Section / Batch:	2024	Date:	1/08/2026

Code of Conduct

I _____ C Prathibha _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ C Prathibha _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

Apply state transitions to represent suffix transformations and irregular forms in verb inflection.

Differentiate between regular and irregular inflectional patterns within the dataset.

Ensure consistent mapping of all forms to a common root representation.

Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

ANSWERS

1. Morphological Analysis Pipeline

Input Words: connected, connecting, connection

Step 1: Decompose into Root and Suffix

Word	Root	Suffix
connected	connect	-ed
connecting	connect	-ing
connection	connect	-ion

Step 2: Identify Word Type

Word	Type	Reason
connected	Inflectional	Indicates past tense
connecting	Inflectional	Indicates present participle
connection	Derivational	Forms a noun from a verb

Step 3: Normalize

All words are normalized to the base form:

connect

Morphological Parsing

connected

└─ Root : connect

└─ Suffix : -ed

connecting

└─ Root : connect

└─ Suffix : -ing

connection

└─ Root : connect

└─ Suffix : -ion

Final Normalized Output

Input	Parsed Structure	Normalized Output
connected	connect + ed	connect
connecting	connect + ing	connect
connection	connect + ion	connect

2. Morphological Parsing for Sentiment Analysis

Input Words: unhappy, happiness, happily

Step 1: Rule-Based Decomposition

Word	Prefix	Root	Suffix
unhappy	un-	happy	—
happiness	—	happy	-ness
happily	—	happy	-ly

Step 2: Semantic Consistency

All words share the same base concept:

happy

Step 3: Classification

Word	Transformation
unhappy	Derivational (negative prefix)
happiness	Derivational (noun formation)
happily	Derivational (adverb formation)

Morphological Breakdown

unhappy

|—— Prefix : un-

|—— Root : happy

happiness

|—— Root : happy

└—— Suffix : -ness

happily

|—— Root : happy

└—— Suffix : -ly

Final Output

Input	Breakdown	Root
unhappy	un + happy	happy
happiness	happy + ness	happy
happily	happy + ly	happy

3. Stemming-Based Preprocessing Module

Input Words: played, player, playing

Step 1: Remove Affixes

Word	Rule Applied	Stem
played	Remove "-ed"	play
player	Remove "-er"	play
playing	Remove "-ing"	play

Step 2: Classification

Word	Type
played	Inflectional
player	Derivational
playing	Inflectional

Morphological Breakdown

played

└─ Root : play

└─ Suffix : -ed

player

└─ Root : play

└─ Suffix : -er

playing

└─ Root : play

└─ Suffix : -ing

Final Output

Input	Stem	Classification
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

4. Finite-State Morphological Parsing

Input Words: writes, writing, written

Finite State Transitions

Start

|



Root (write)

|

└─ +s → writes

|

└─ +ing → writing

|

└─ Irregular +en → written

Morphological Breakdown

Word	Root	Suffix	Pattern
writes	write	-s	Regular
writing	write	-ing	Regular
written	write	-en	Irregular

Classification

Word	Type
writes	Regular Inflection
writing	Regular Inflection
written	Irregular Inflection

Normalized Output

All forms are mapped to:

write

Final Output Table

Input	State Transition	Morphological Breakdown	Normalized Output
writes	write → +s	write + s	write
writing	write → +ing	write + ing	write
written	write → +en	write + en	write

5. Porter Stemmer-Based Preprocessing

Input Words: relational, relation, relate

Porter Stemming Steps

Word 1: relational

relational

↓ Remove -al

relation

↓ Remove -ion

relat

Final Stem: relat

Word 2: relation

relation

↓ Remove -ion

relat

Final Stem: relat

Word 3: relate

relate

↓ Remove final -e

relat

Final Stem: relat

Intermediate Forms

Word	Step 1	Step 2	Final Stem
relational	relation	relat	relat
relation	relat	—	relat
relate	relat	—	relat

Final Output

Input	Stemming Steps	Final Stem
relational	relational → relation → relat	relat
relation	relation → relat	relat
relate	relate → relat	relat

Summary

Question	Common Base/Stem
connected, connecting, connection	connect
unhappy, happiness, happily	happy
played, player, playing	play
writes, writing, written	write
relational, relation, relate	relat (Porter Stemmer output)